

Optimal size at flowering

Loïc Pages

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```
IPM_data <- read.csv("newdata.csv")

MatrixDim <- 50
RecrRate <- 0.338 #recruitment rate

#Forcer l'age maximal à 8
AgeMax <- 8
IPM_data$Age[IPM_data$Age > AgeMax] <- AgeMax

centauree_data <- IPM_data[!is.na(IPM_data$Size0Mars) & !is.na(IPM_data$Age),]
flower_data <- IPM_data[IPM_data$Flowering==1,] %>% filter(year!=1994)
```

Données climatique (température et jours de pluie)

```
climate_data <- read_csv("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Code/IPM/1")
```

```
## Rows: 12053 Columns: 8
## -- Column specification -----
## Delimiter: ","
## dbl (8): AN, MOIS, JOUR, precipitations, T moyenne, T min,, T max,, Prec > 1 mm
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
climate_data <- climate_data %>%
  rename(
    "Tmean" = "T moyenne",
    "Tmin" = "T min,",
    "Tmax" = "T max,",
    "preci1mm" = "Prec > 1 mm",
    "year" = "AN"
  ) %>%
  filter(!year<1994)

climate_data_mean <- climate_data %>%
  group_by(year) %>%
  summarise(Tmean = mean(Tmean),
            NbWetDays = sum(preci1mm),
            Tmax = max(Tmax))
```

```

climate_data_season <- climate_data %>%
  group_by(year) %>%
  add_column(Season=NA)
climate_data_season[climate_data_season$MOIS %in% c(1,2,12),]$Season <- "Winter"
climate_data_season[climate_data_season$MOIS %in% c(3,4,5),]$Season <- "Spring"
climate_data_season[climate_data_season$MOIS %in% c(6,7,8),]$Season <- "Summer"
climate_data_season[climate_data_season$MOIS %in% c(9,10,11),]$Season <- "Autumn"
climate_data_season <- climate_data_season %>%
  group_by(year,Season) %>%
  summarise(Tmean = mean(Tmean),
            NbWetDays = sum(preci1mm),
            Tmax = max(Tmax))

```

'summarise()' has grouped output by 'year'. You can override using the
'.groups' argument.

```

climate_data_month <- climate_data %>%
  group_by(year,MOIS) %>%
  summarise(Tmean = mean(Tmean),
            NbWetDays = sum(preci1mm),
            Tmax = max(Tmax))

```

'summarise()' has grouped output by 'year'. You can override using the
'.groups' argument.

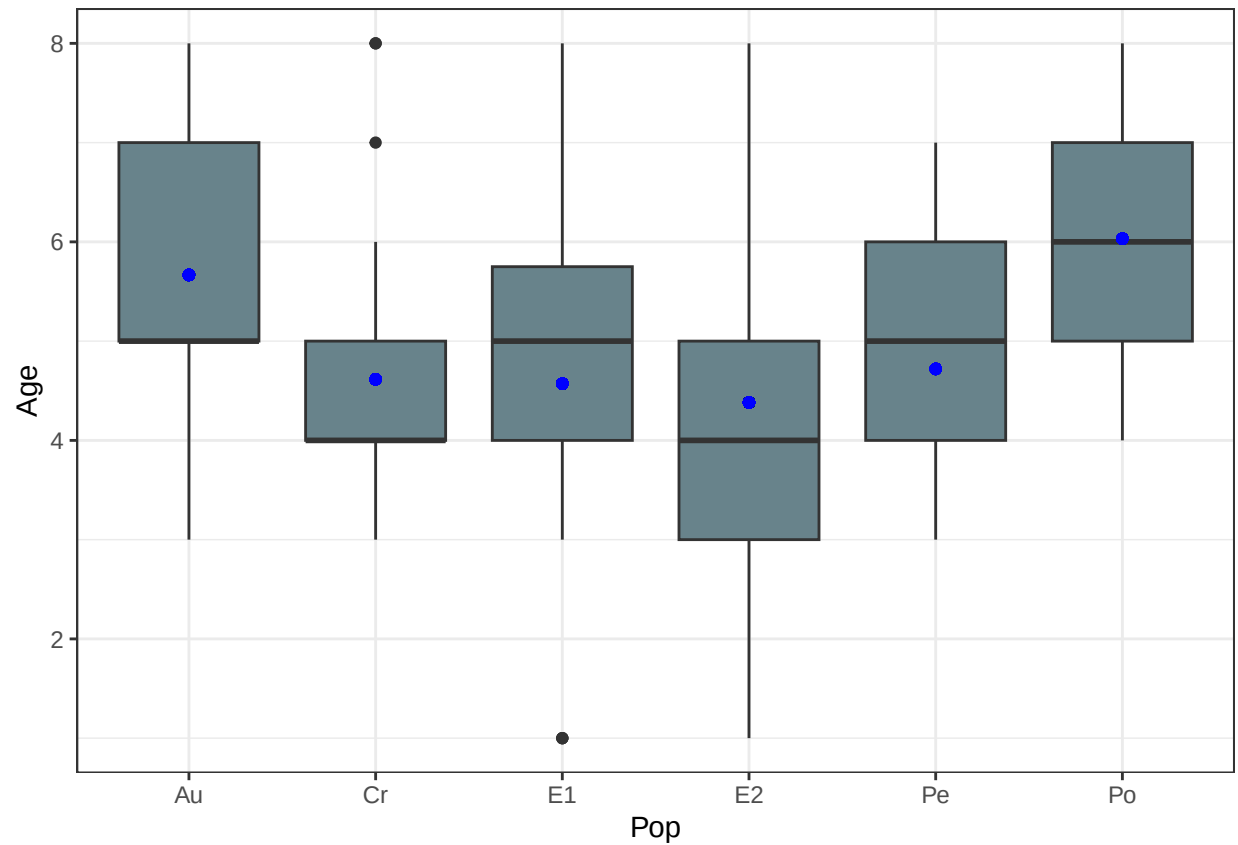
Observed data

Observed mean age at flowering

```

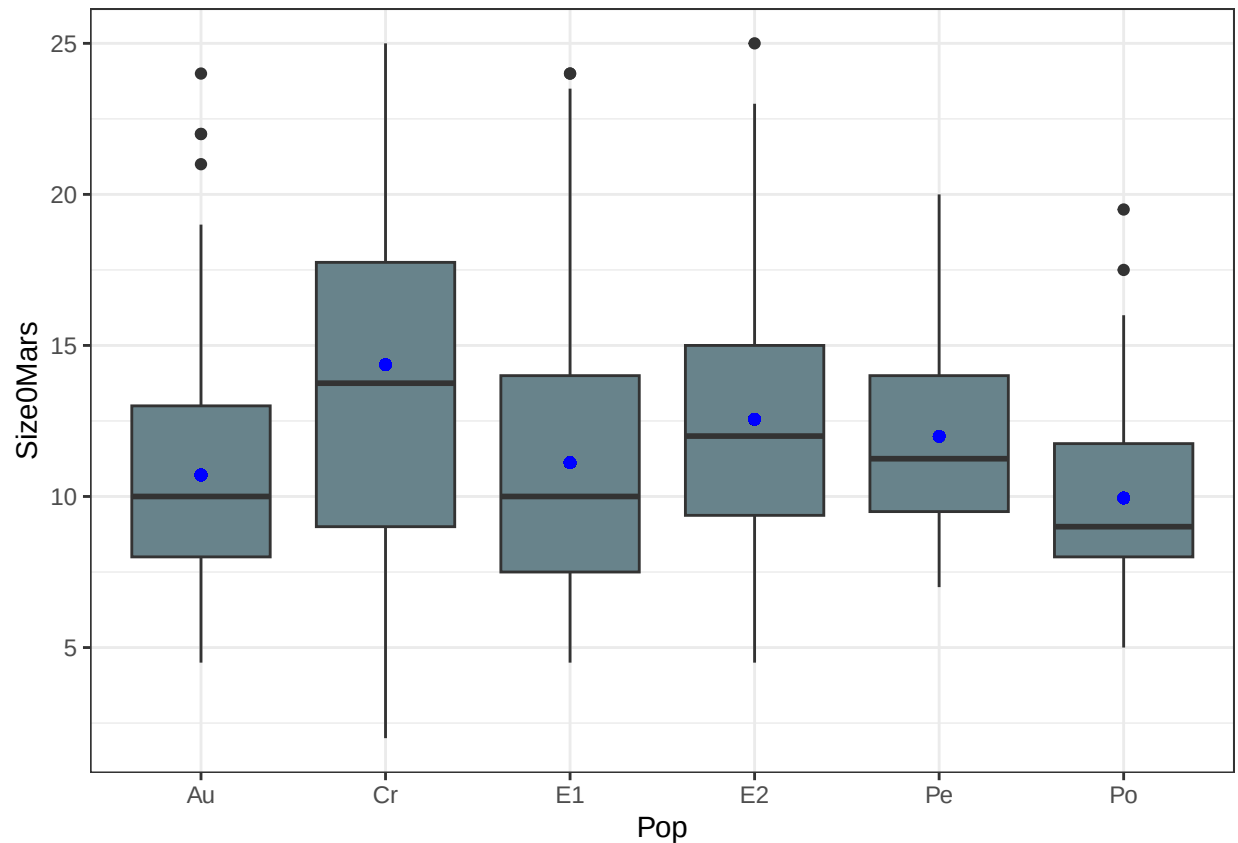
#Observed
flower_data %>%
  group_by(Pop) %>%
  mutate(mean.age = mean(Age,na.rm=TRUE)) %>%
  ggplot(aes(x=Pop, y=Age)) +
  geom_boxplot(fill="lightblue4")+
  geom_point(aes(y=mean.age),color= "blue")+
  theme_bw()

```



Observed mean size at flowering

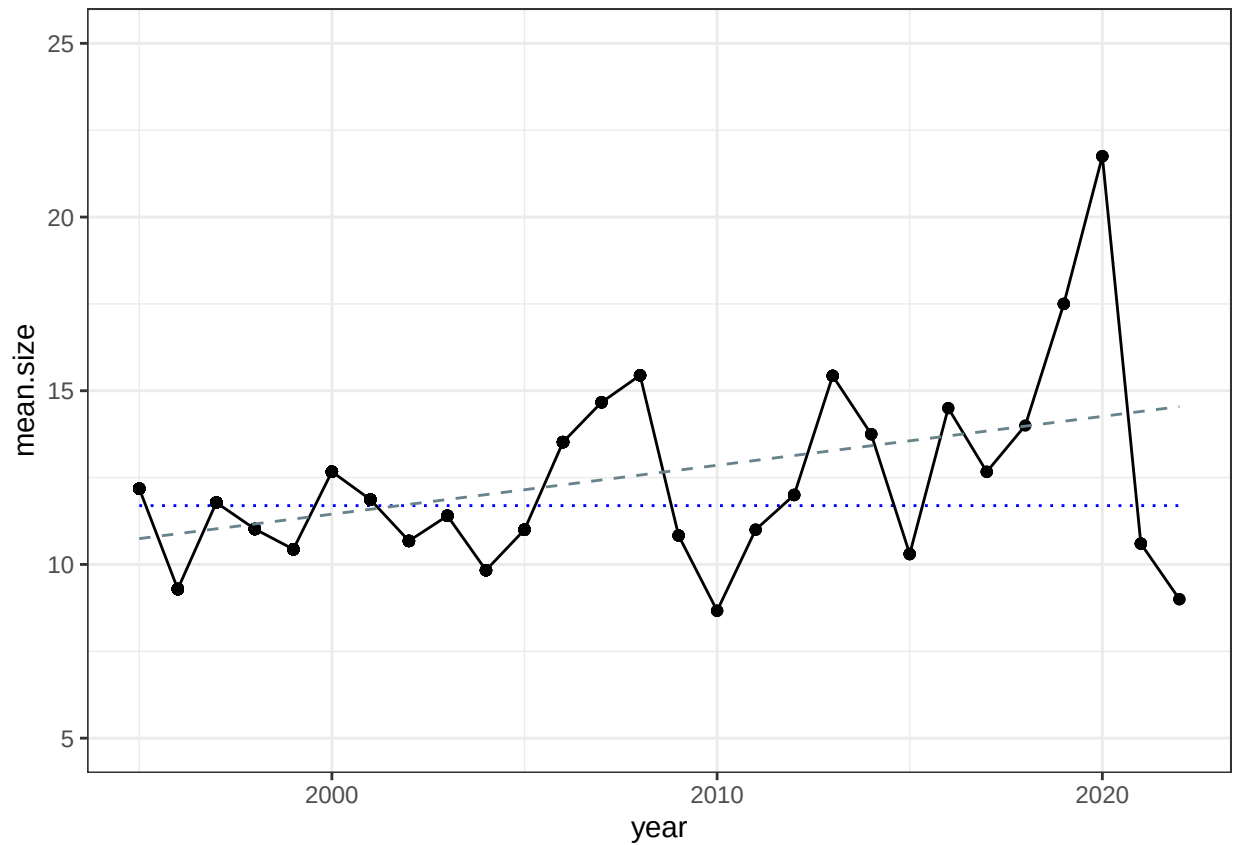
```
flower_data %>%
  group_by(Pop) %>%
  mutate(mean.size = mean(Size0Mars,na.rm=TRUE)) %>%
  ggplot(aes(x=Pop, y=Size0Mars)) +
  geom_boxplot(fill="lightblue4")+
  geom_point(aes(y=mean.size),color= "blue")+
  theme_bw()
```



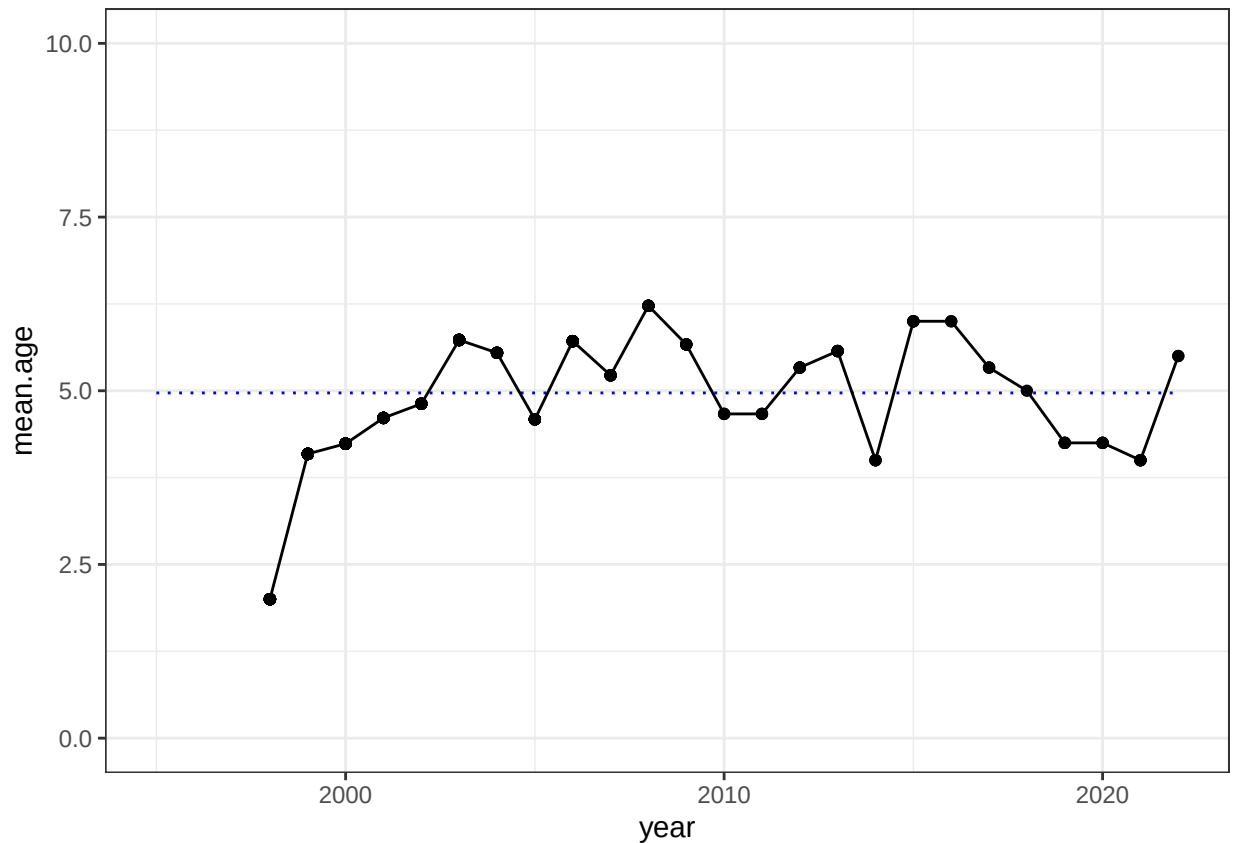
Temporal trend of mean size and age at flowering

```
flower_data %>%
  mutate(meanAll = mean(Size0Mars,na.rm=TRUE)) %>%
  group_by(year) %>%
  summarize(mean.size = mean(Size0Mars,na.rm=TRUE),
            meanAll = meanAll) %>%
  ungroup(year) %>%
  mutate(lm = predict(lm(mean.size~year))) %>%
  ggplot(aes(x=year, y=mean.size)) +
  geom_line(aes(y=meanAll),col="blue", linetype=3)+
  geom_point()+
  geom_line()+
  geom_line(aes(y=lm),col="lightblue4", linetype=2)+
  theme_bw()+
  ylim(5,25)
```

'summarise()' has grouped output by 'year'. You can override using the
'.groups' argument.



```
flower_data %>%  
  mutate(meanAll = mean(Age,na.rm=TRUE)) %>%  
  group_by(year) %>%  
  mutate(mean.age = mean(Age,na.rm=TRUE)) %>%  
  ggplot(aes(x=year, y=mean.age)) +  
  geom_line(aes(y=meanAll),col="blue", linetype=3)+  
  geom_point()+  
  geom_line()+  
  theme_bw()+  
  ylim(0,10)
```

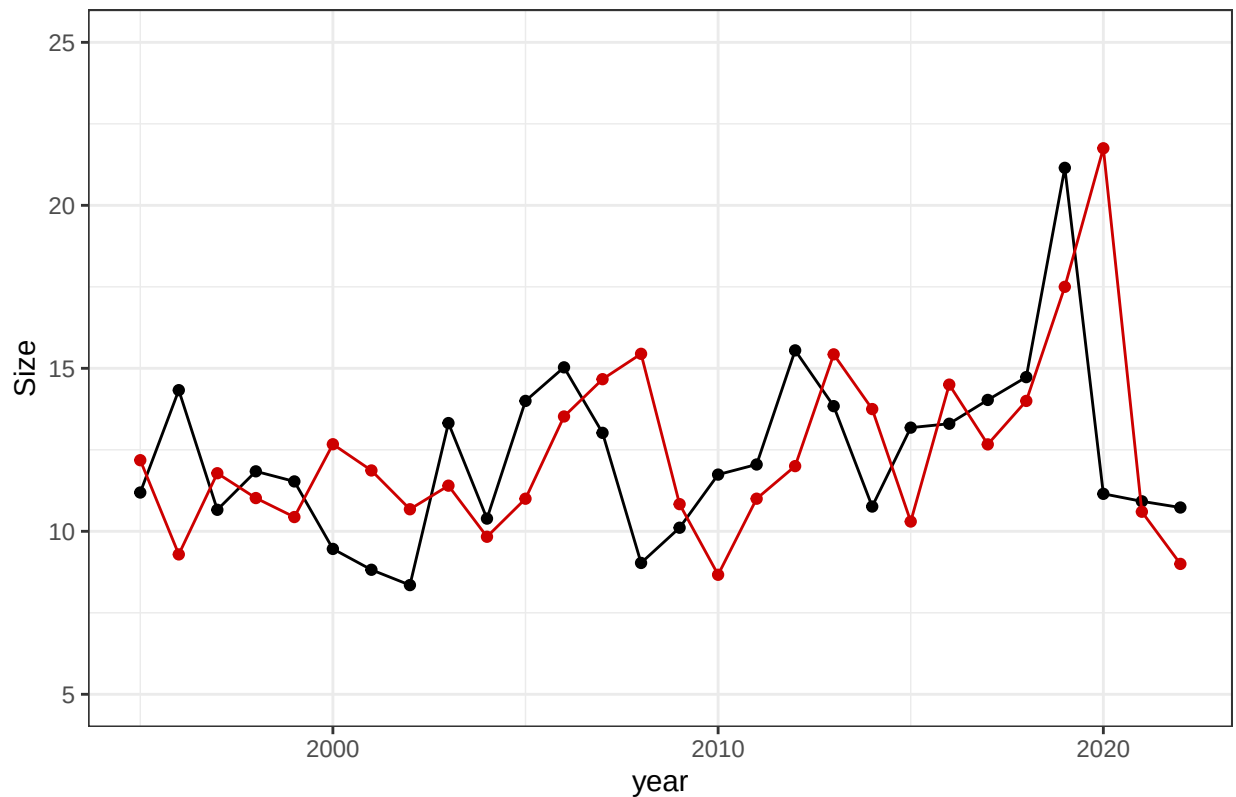


Optimal size at flowering comparing to observed size

Optimal size at flowering for each year - the model consider that for each year, if we stay in this year indefinitely, the optimal flowering size is `opti.size`.

```
optimalsize_data %>%
  ggplot(aes(x=year)) +
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=opti.size))+
  geom_point(aes(y=mean.size),col="red3")+
  geom_line(aes(y=mean.size),col="red3")+
  # geom_point(aes(y=predict.size),col="blue")+
  # geom_line(aes(y=predict.size),col="blue")+
  theme_bw()+
  labs(title="Size at flowering. black : optimal predicted, red : observed mean")+
  ylab("Size")+
  ylim(5,25)
```

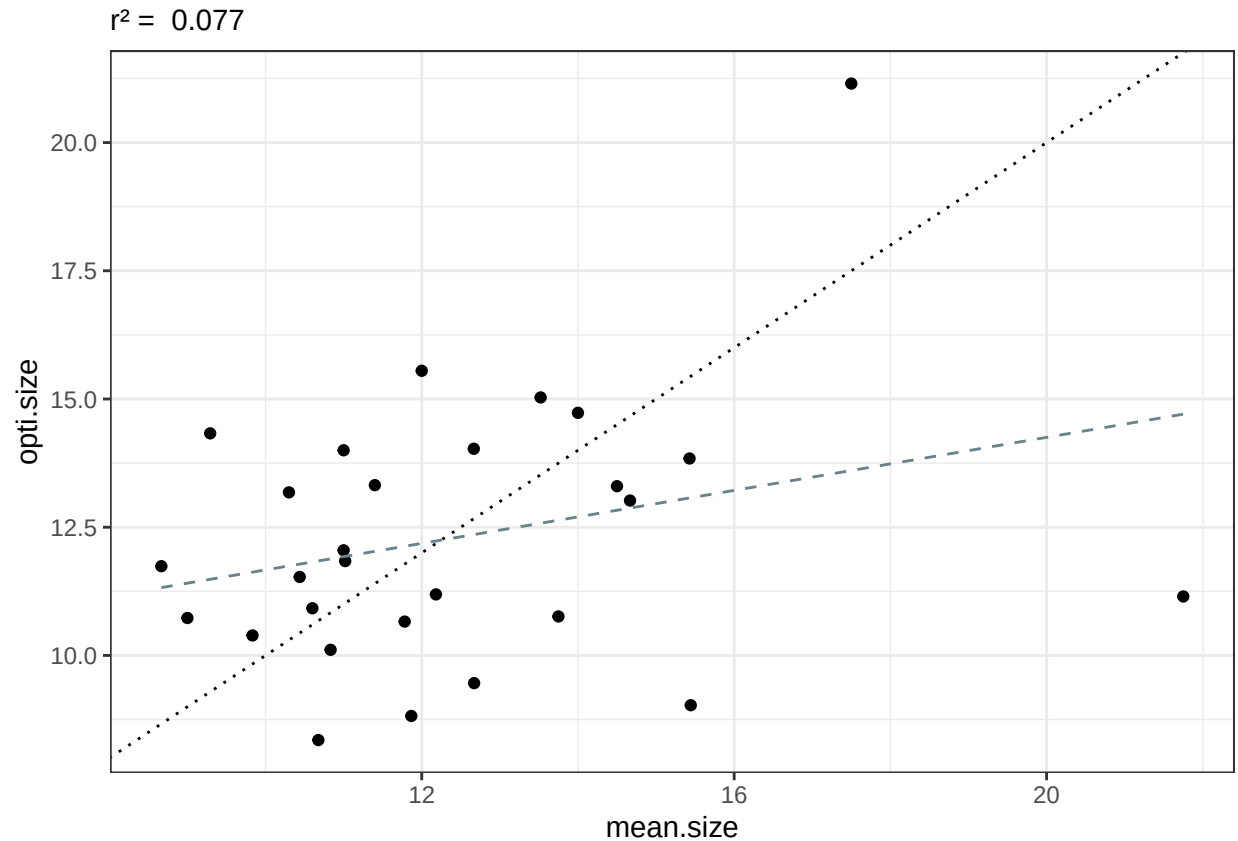
Size at flowering. black : optimal predicted, red : observed mean



Optimal size at flowering compared to observed mean size at flowering

```
r2 <- optimalsize_data %>%
  mutate(lm = predict(lm(opti.size~mean.size))) %>%
  summarize(rsquared = summary(lm(opti.size~mean.size))$r.squared)

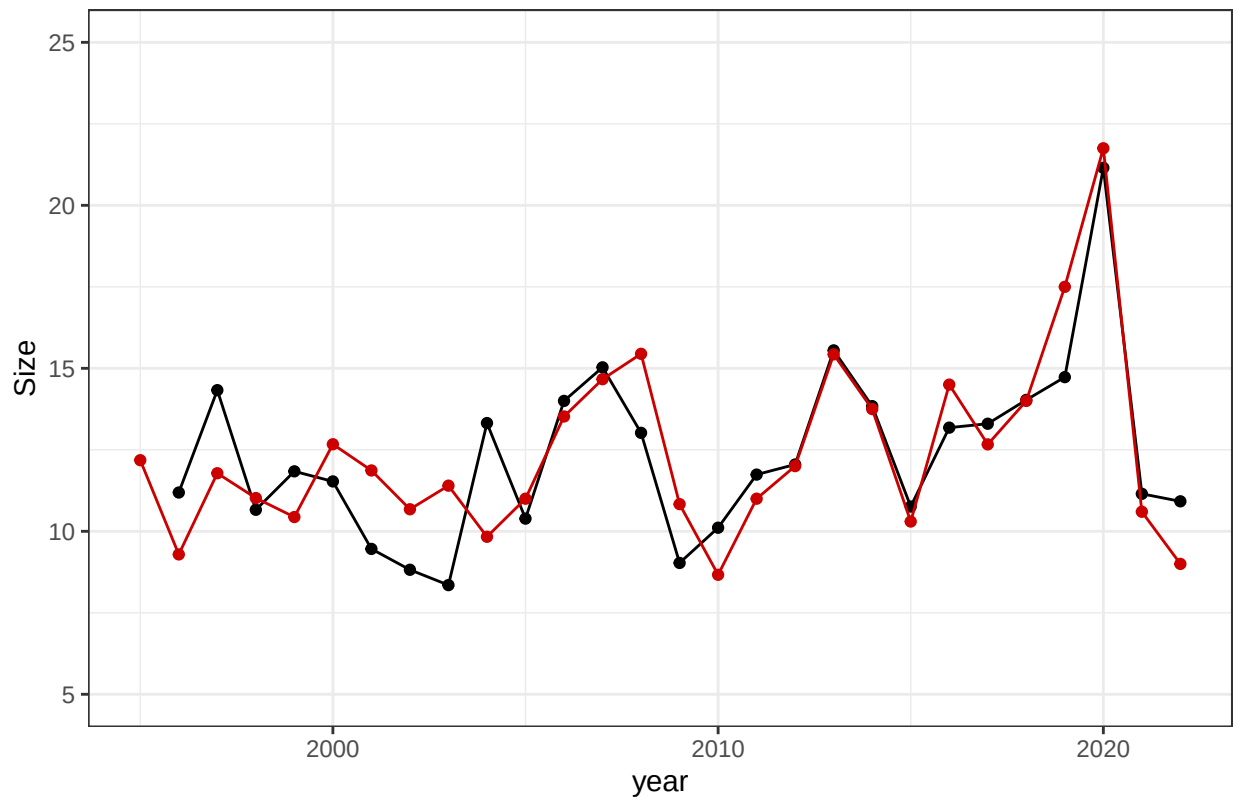
optimalsize_data %>%
  mutate(lm = predict(lm(opti.size~mean.size))) %>%
  ggplot(aes(x=mean.size, y=opti.size)) +
  geom_point()+
  geom_abline(intercept=0,slope=1,linetype=3)+
  geom_line(aes(y=lm), linetype=2, col="lightblue4")+
  theme_bw()+
  labs(subtitle = str_c("r2 = ",round(r2,3)))
```



Optimal size at flowering for each year. If we assume that it is not the current year but the previous year which impacts the choice of flowering, so we compare the mean size at flowering at year t to the optimal size at year $t-1$.

```
optimalsize_data %>%
  mutate(opti.size = c(NA, opti.size[-28])) %>%
  ggplot(aes(x=year)) +
  geom_point(aes(y=opti.size)) +
  geom_line(aes(y=opti.size)) +
  geom_point(aes(y=mean.size), col="red3") +
  geom_line(aes(y=mean.size), col="red3") +
  # geom_point(aes(y=predict.size), col="blue") +
  # geom_line(aes(y=predict.size), col="blue") +
  theme_bw() +
  labs(title="Size at flowering. black : optimal predicted, red : observed mean") +
  ylab("Size") +
  ylim(5, 25)
```

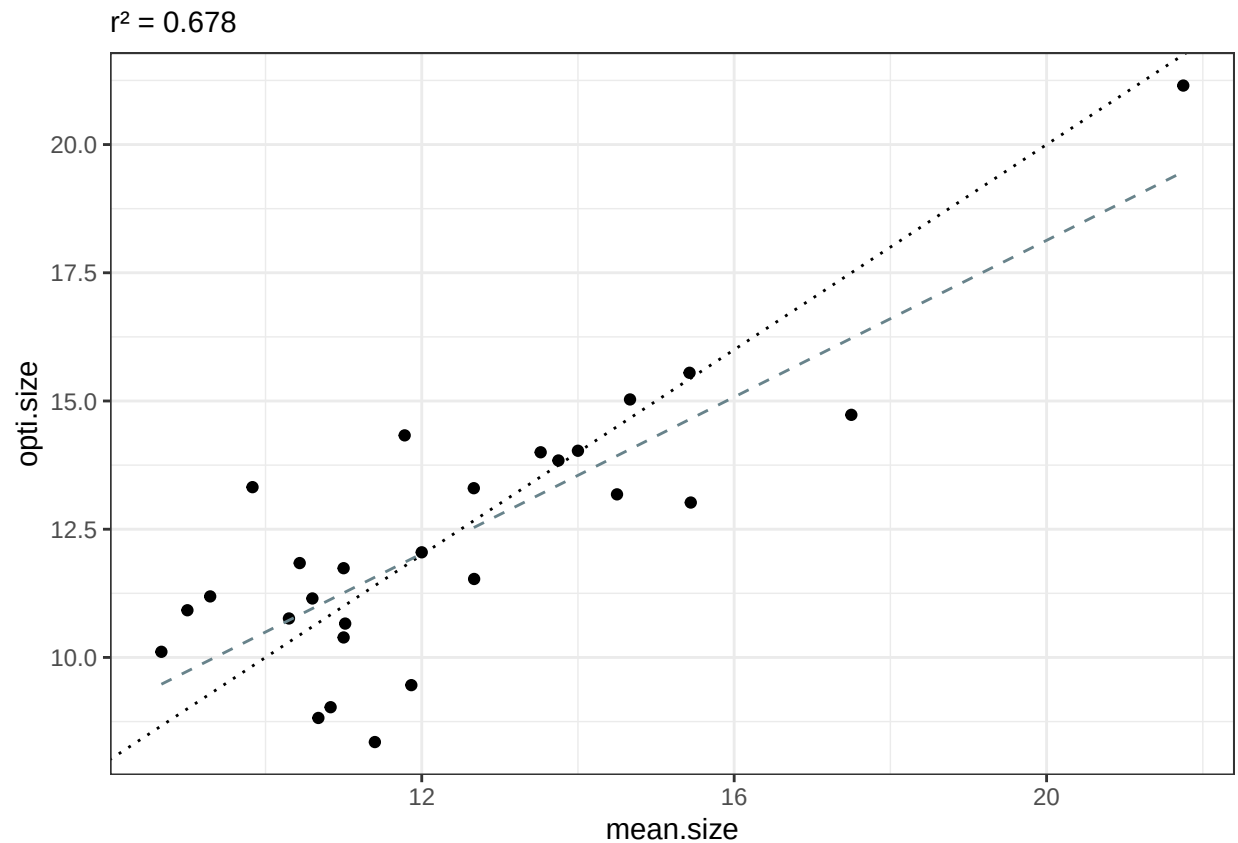

Size at flowering. black : optimal predicted, red : observed mean



Optimal size at flowering compared to observed mean size at flowering

```
r2 <- optimalsize_data %>%
  mutate(opti.size = c(NA,opti.size[-28]),
         lm = c(NA,predict(lm(opti.size~mean.size)))) %>%
  summarize(rsquared = summary(lm(opti.size~mean.size))$r.squared)

optimalsize_data %>%
  mutate(opti.size = c(NA,opti.size[-28]),
         lm = c(NA,predict(lm(opti.size~mean.size))),
         rsquared = rep(summary(lm(opti.size~mean.size))$r.squared,28)) %>%
  ggplot(aes(x=mean.size, y=opti.size)) +
  geom_point()+
  geom_abline(intercept=0,slope=1,linetype=3)+
  geom_line(aes(y=lm), linetype=2, col="lightblue4")+
  theme_bw()+
  labs(subtitle = str_c("r2 = ",as.character(round(r2,3))))
```

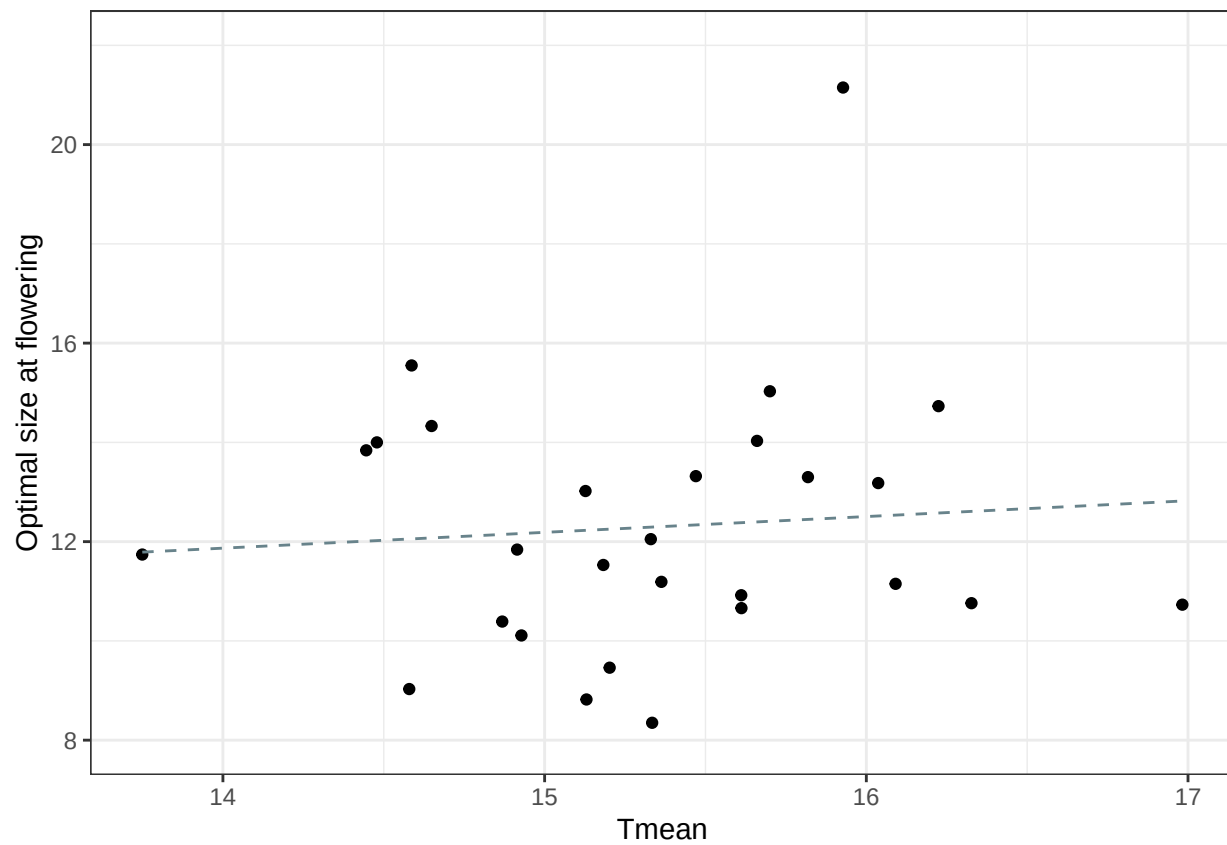


Climate

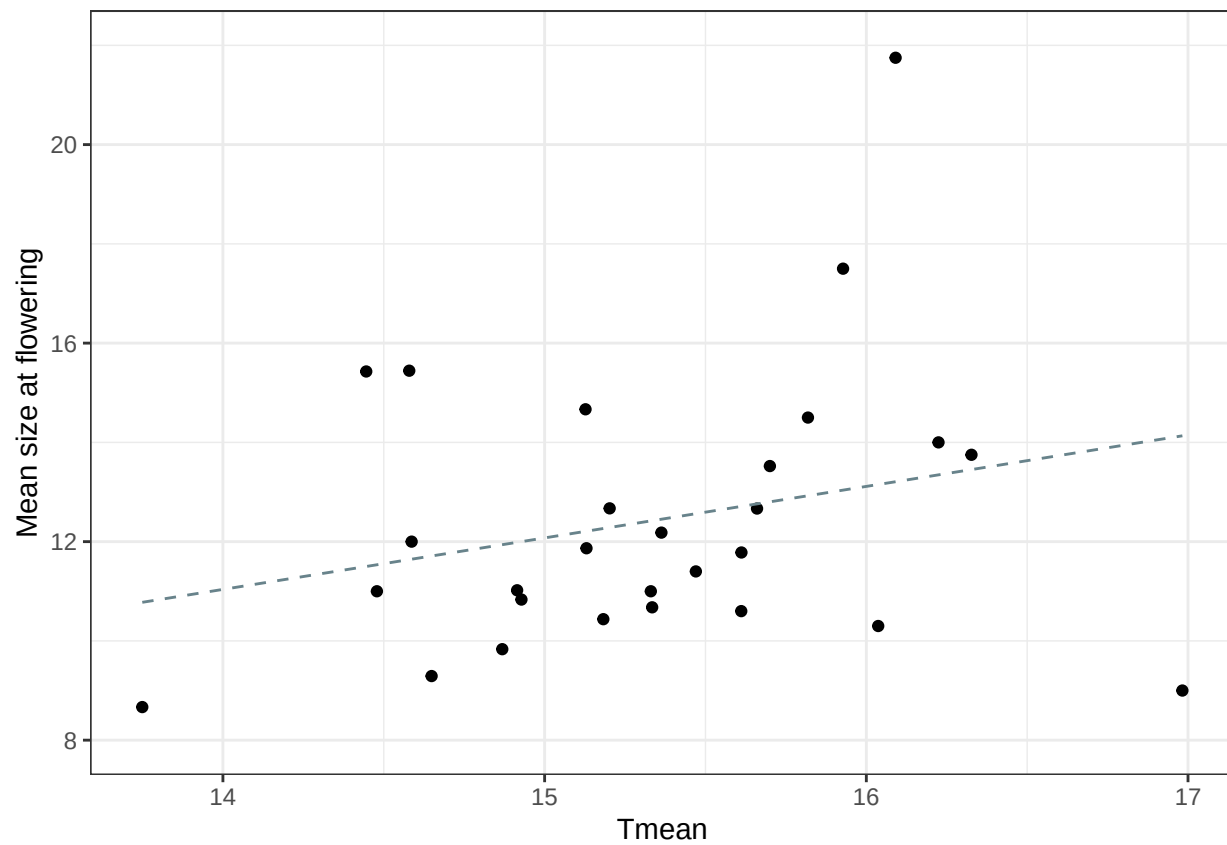
Comparison between size at flowering and mean year temperature.

```
flowclimate_data <- left_join(climate_data_mean, optimalsize_data, by="year") %>% filter(year!=1994)

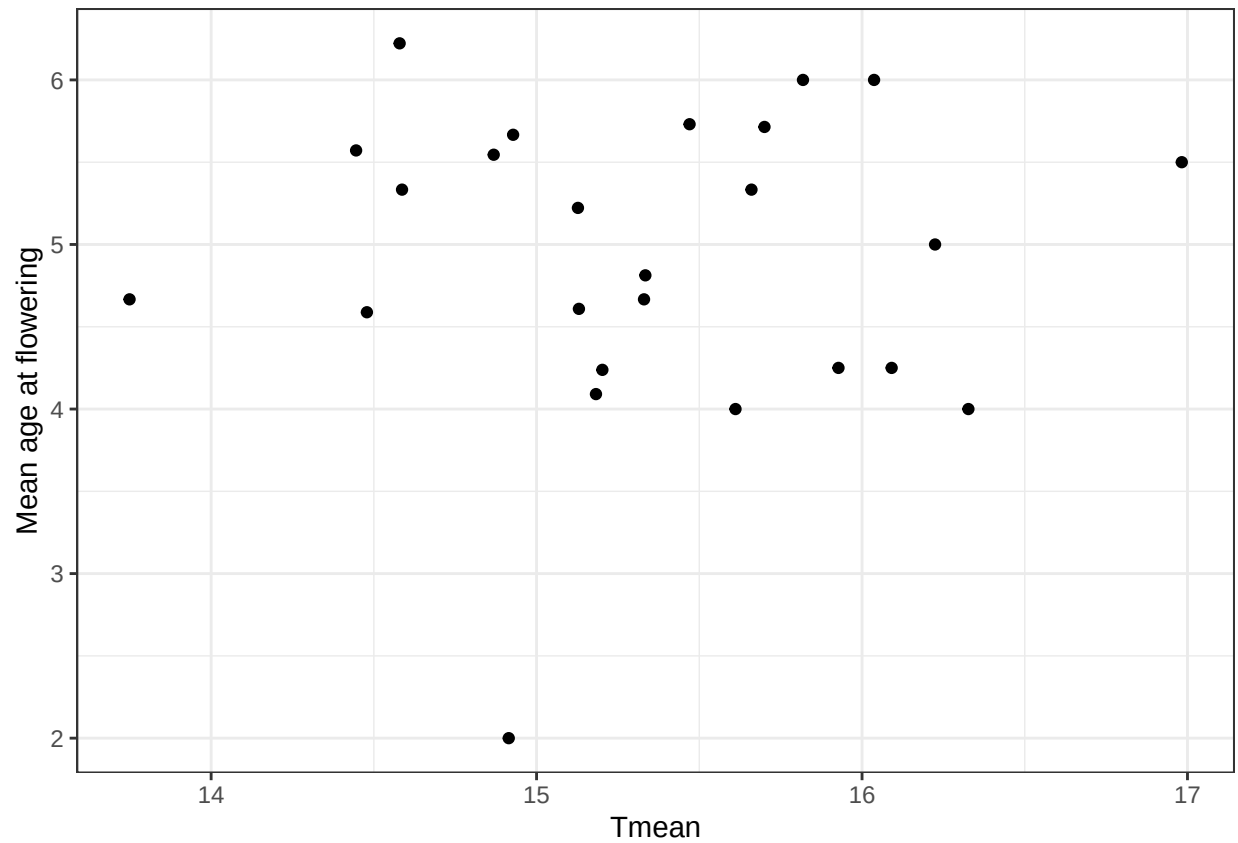
flowclimate_data %>%
  ggplot(aes(x=Tmean)) +
  geom_point(aes(y=opti.size)) +
  geom_line(aes(y=predict(lm(opti.size~Tmean, data=flowclimate_data))), linetype=2, col="lightblue4") +
  theme_bw() +
  labs(y="Optimal size at flowering") +
  ylim(8,22)
```



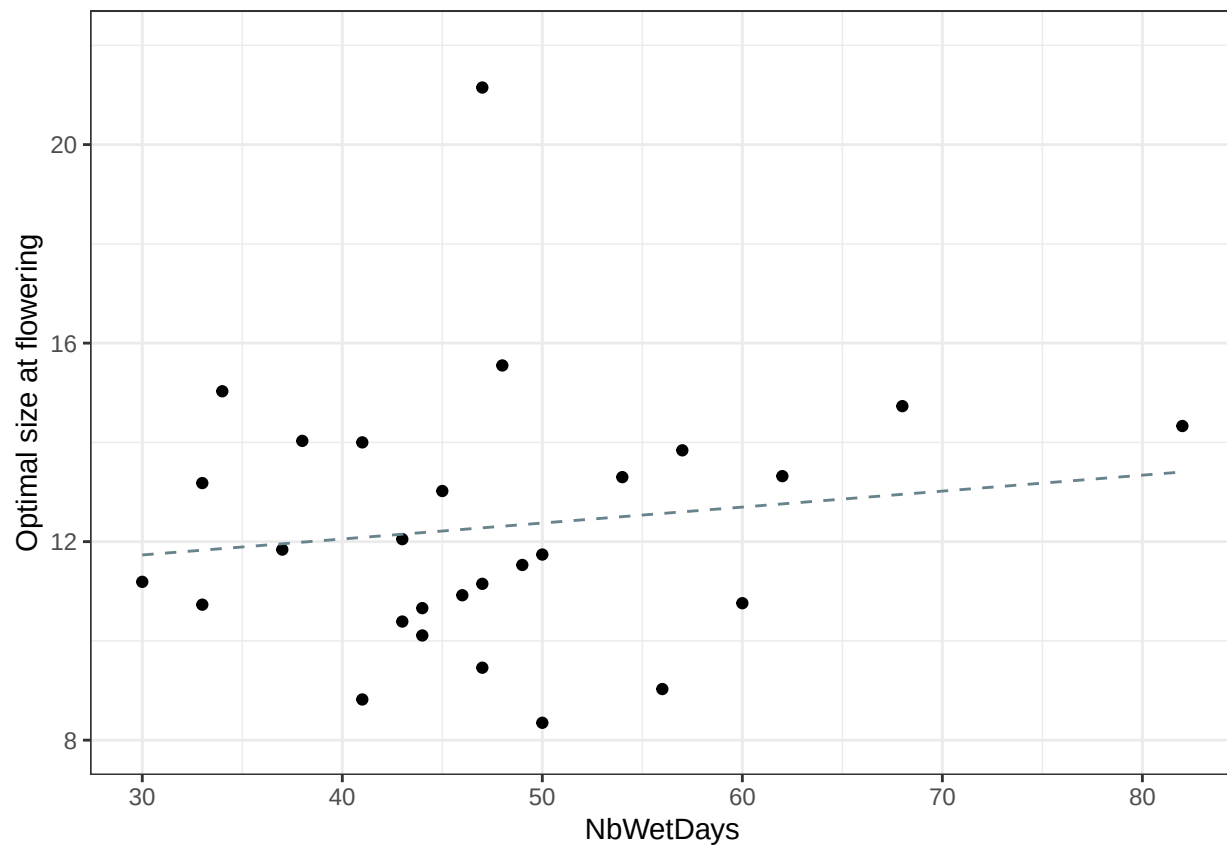
```
flowclimate_data %>%
  ggplot(aes(x=Tmean)) +
  geom_point(aes(y=mean.size)) +
  geom_line(aes(y=predict(lm(mean.size~Tmean,data=flowclimate_data))), linetype=2, col="lightblue4") +
  theme_bw() +
  labs(y="Mean size at flowering") +
  ylim(8,22)
```



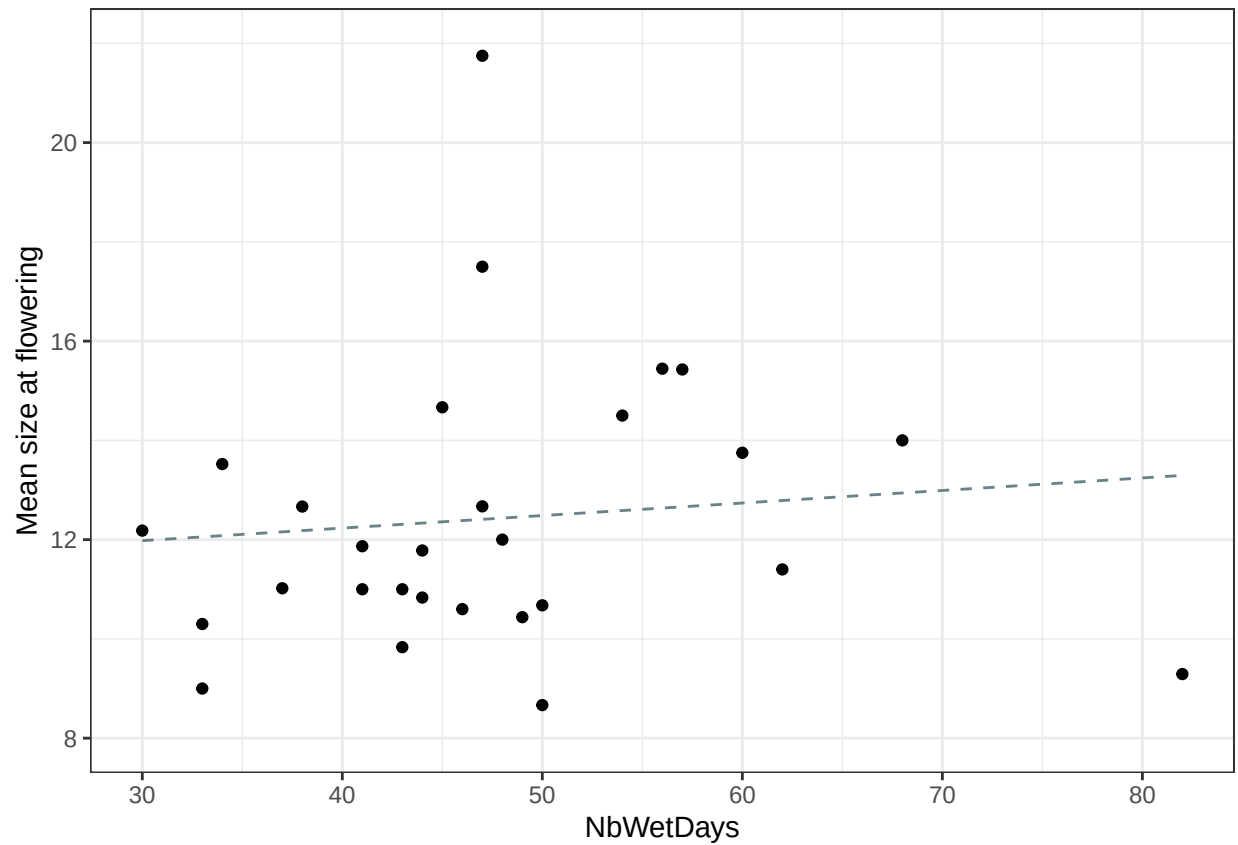
```
flowclimate_data %>%  
  ggplot(aes(x=Tmean)) +  
  geom_point(aes(y=mean.age))+  
  theme_bw()+  
  labs(y="Mean age at flowering")
```



```
flowclimate_data %>%
  ggplot(aes(x=NbWetDays)) +
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=predict(lm(opti.size~NbWetDays,data=flowclimate_data))), linetype=2, col="lightblue4"),
  theme_bw()+
  labs(y="Optimal size at flowering")+
  ylim(8,22)
```



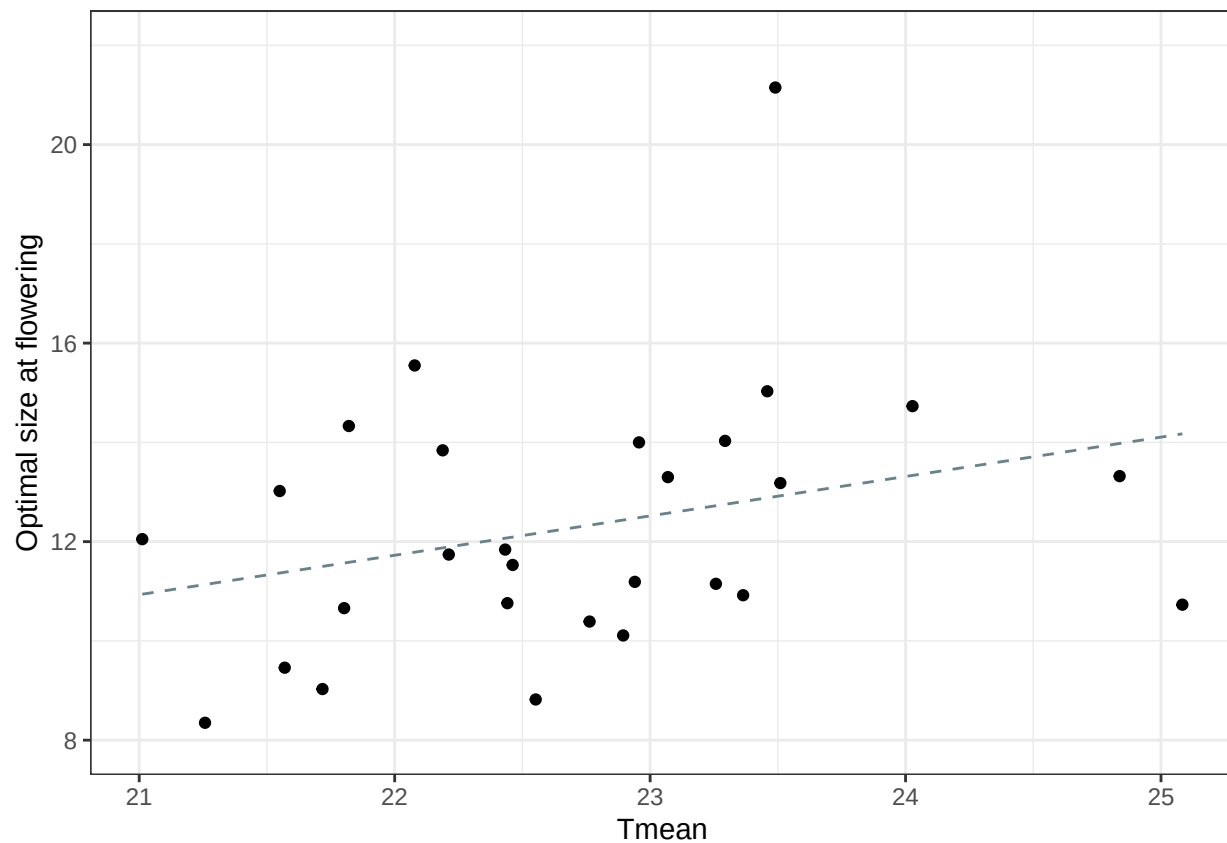
```
flowclimate_data %>%
  ggplot(aes(x=NbWetDays)) +
  geom_point(aes(y=mean.size)) +
  geom_line(aes(y=predict(lm(mean.size~NbWetDays,data=flowclimate_data))), linetype=2, col="lightblue4")
  theme_bw() +
  labs(y="Mean size at flowering") +
  ylim(8,22)
```



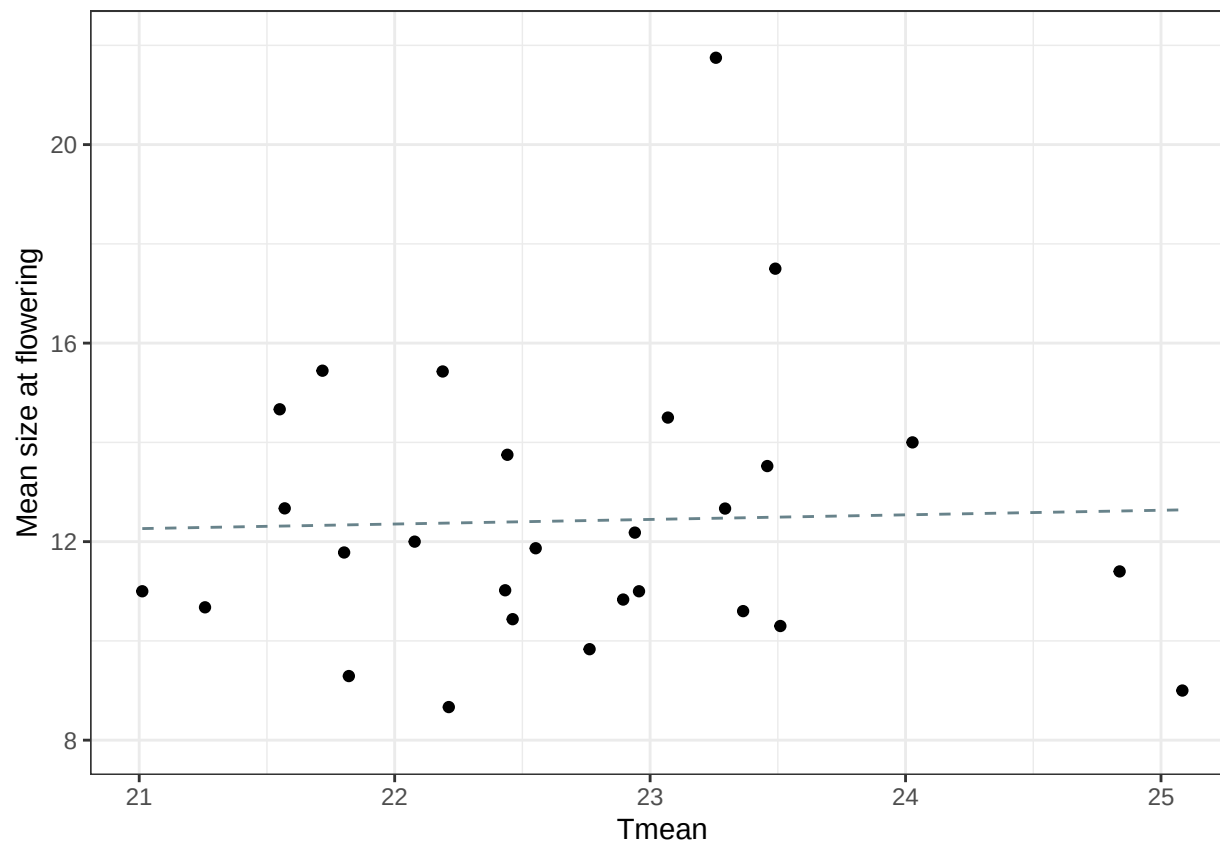
Comparison between size at flowering and mean summer temperature.

```
flowclimate_data_season <- left_join(climate_data_season, optimalsize_data, by="year") %>% filter(year!=)

flowclimate_data_season %>%
  filter(Season=="Summer") %>%
  ggplot(aes(x=Tmean)) +
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=predict(lm(opti.size~Tmean,data=flowclimate_data_season%>%
    filter(Season=="Summer"))), linetype=2, col="lightblue4")+
  theme_bw()+
  labs(y="Optimal size at flowering")+
  ylim(8,22)
```



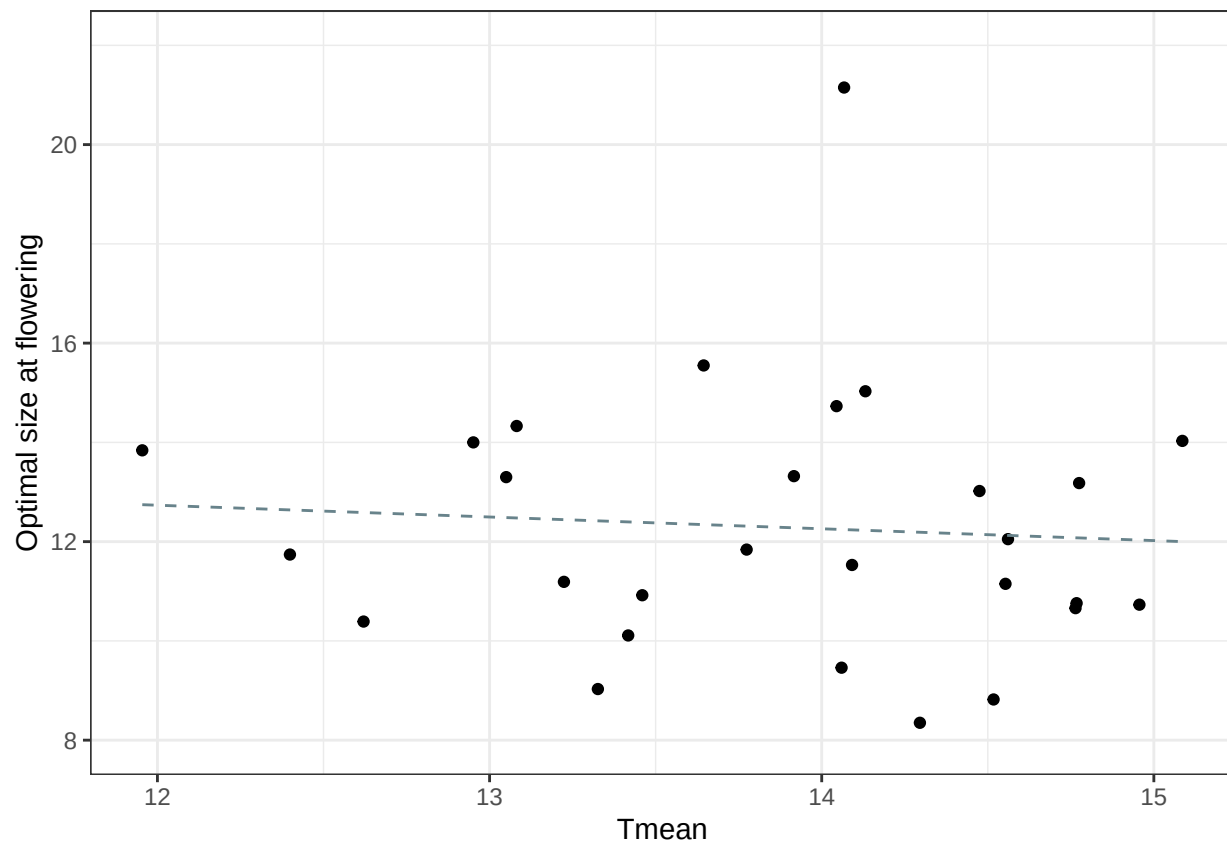
```
flowclimate_data_season %>%
  filter(Season=="Summer") %>%
  ggplot(aes(x=Tmean)) +
  geom_point(aes(y=mean.size))+
  geom_line(aes(y=predict(lm(mean.size~Tmean,data=flowclimate_data_season%>%
    filter(Season=="Summer")))), linetype=2, col="lightblue4")+
  theme_bw()+
  labs(y="Mean size at flowering")+
  ylim(8,22)
```

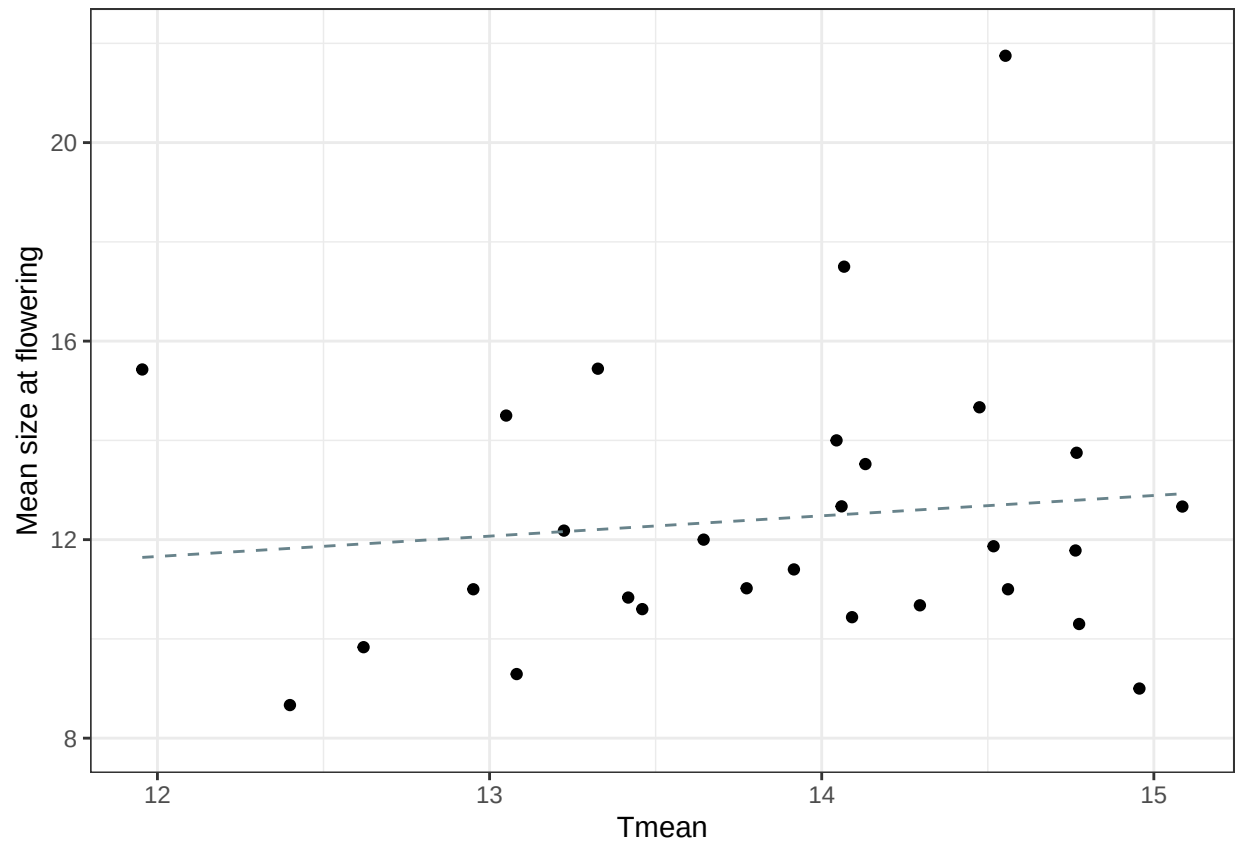
Comparison between size at flowering and mean spring temperature.

```
flowclimate_data_season <- left_join(climate_data_season, optimalsize_data, by="year") %>% filter(year!=2015)

flowclimate_data_season %>%
  filter(Season=="Spring") %>%
  ggplot(aes(x=Tmean)) +
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=predict(lm(opti.size~Tmean,data=flowclimate_data_season%>%
    filter(Season=="Spring"))), linetype=2, col="lightblue4")+
  theme_bw()+
  labs(y="Optimal size at flowering")+
  ylim(8,22)
```



```
flowclimate_data_season %>%
  filter(Season=="Spring") %>%
  ggplot(aes(x=Tmean)) +
  geom_point(aes(y=mean.size))+
  geom_line(aes(y=predict(lm(mean.size~Tmean,data=flowclimate_data_season%>%
    filter(Season=="Spring")))), linetype=2, col="lightblue4")+
  theme_bw()+
  labs(y="Mean size at flowering")+
  ylim(8,22)
```



Comparison with other life history traits

```
flowclimate_data2 <- flowclimate_data %>%
  left_join(IPM_data %>%
    group_by(year) %>%
    summarize(mean.surv = mean(Survie))) %>%
  left_join(flower_data %>%
    group_by(year) %>%
    summarize(mean.cap = mean(ifelse(is.na(Capitule), exp(2.31070+0.06846*Size0Mars), Capitule)))) %>%
  left_join(IPM_data %>%
    filter(Age==1) %>%
    group_by(year) %>%
    summarize(Seedlings = sum(Age)))
```

```
## Joining with 'by = join_by(year)'
## Joining with 'by = join_by(year)'
## Joining with 'by = join_by(year)'
```

```
#
# cctl <- flower_data %>%
#   mutate(Capitule = ifelse(is.na(Capitule), exp(2.31070+0.06846*Size0Mars), Capitule)) %>%
#   group_by(Quadrat, year, Pop) %>%
#   summarize(Capitule = sum(Capitule))
```

```

# # cctl$year <- cctl$year+1
# plt <- IPM_data %>%
#   filter(Age==1) %>%
#   group_by(Quadrat,year,Pop) %>%
#   summarize(Seedlings = sum(Age))
# Estb <- left_join(plt, cctl, by = join_by(Quadrat, year, Pop))
#
# flowclimate_data2 <- flowclimate_data2 %>%
#   left_join(Estb %>%
#     group_by(year) %>%
#     summarise(mean.seedling = mean(Seedlings,na.rm=TRUE),
#               mean.cap = mean(Capitule,na.rm=TRUE)))

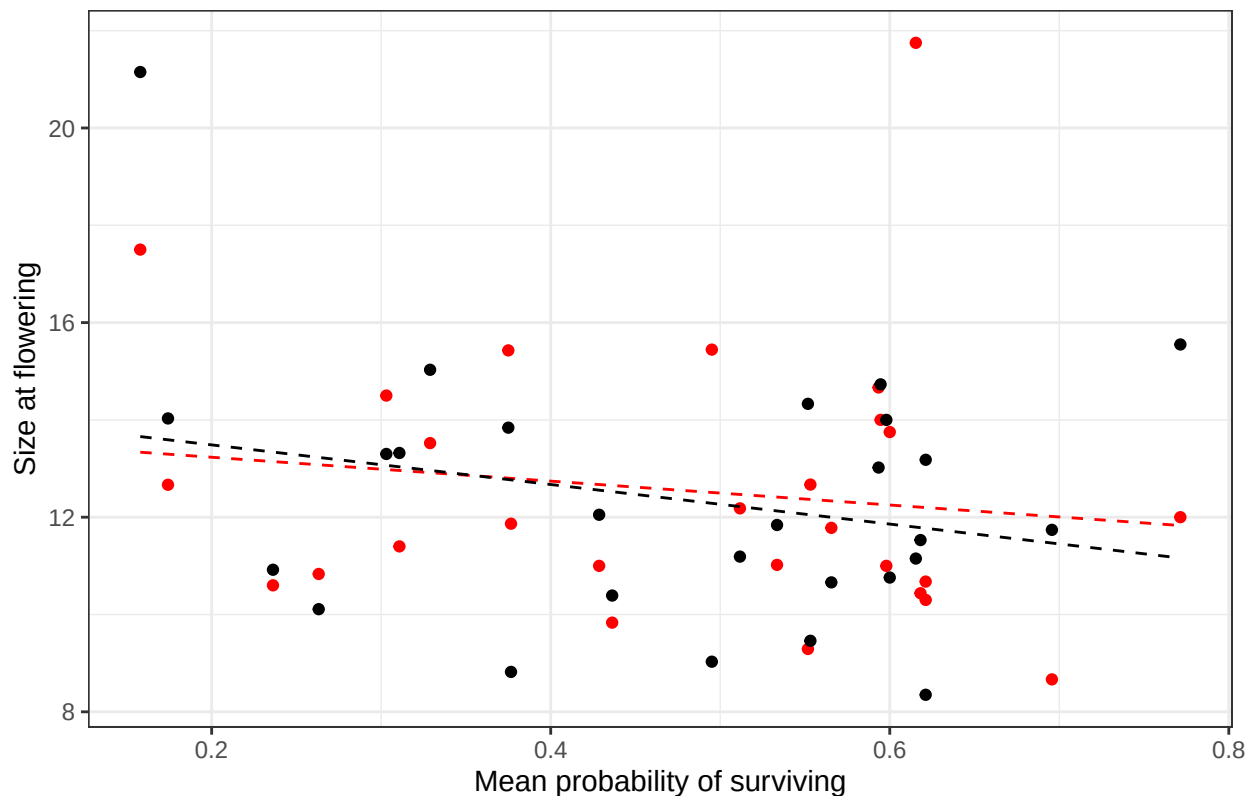
```

```

flowclimate_data2 %>%
  filter(year!=2022) %>%
  ggplot(aes(x=mean.surv))+
  geom_point(aes(y=mean.size),col="red")+
  geom_line(aes(y=predict(lm(mean.size~mean.surv,data=flowclimate_data2))),col="red",linetype=2)+
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=predict(lm(opti.size~mean.surv,data=flowclimate_data2))),linetype=2)+
  theme_bw()+
  labs(title = "red : observed mean size, black : optimal size")+
  ylab("Size at flowering")+
  xlab("Mean probability of surviving")

```

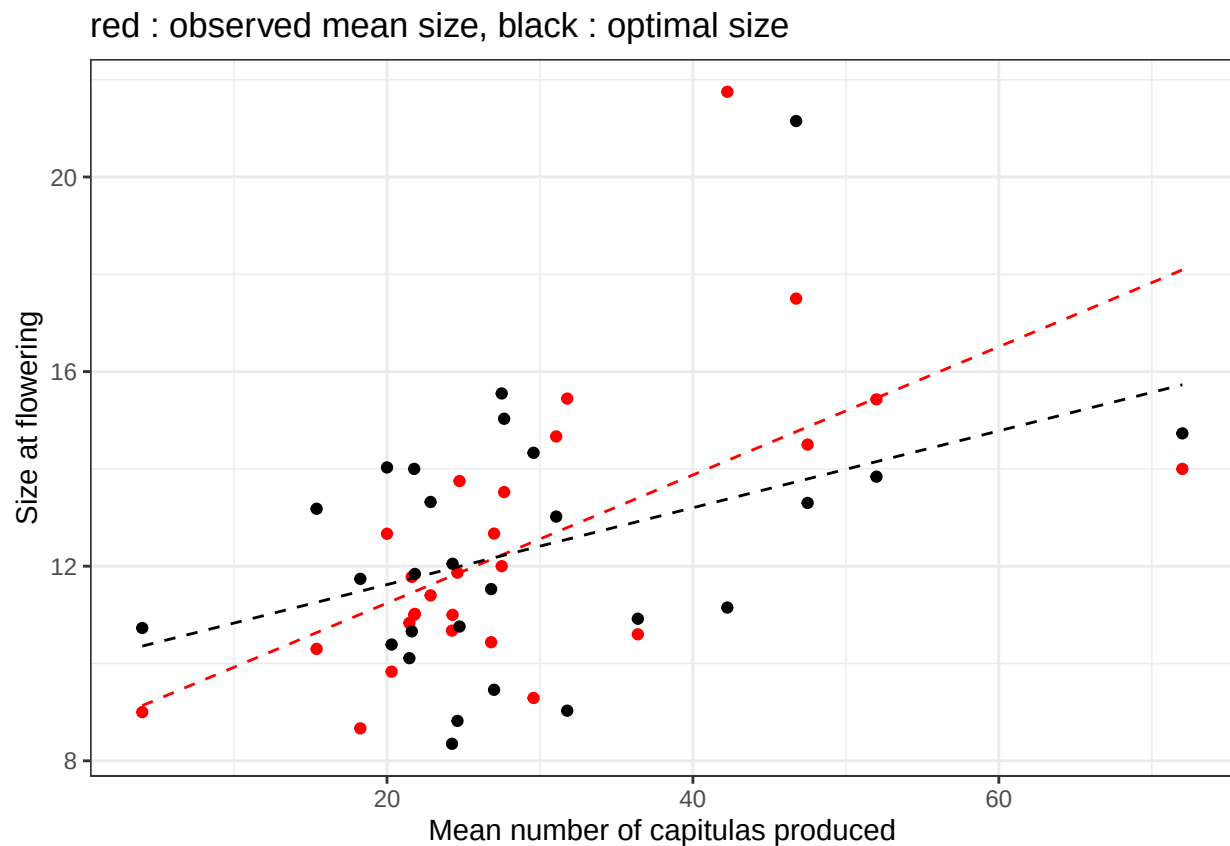
red : observed mean size, black : optimal size



```

flowclimate_data2 %>%
  filter(year!=1995) %>%
  ggplot(aes(x=mean.cap))+
  geom_point(aes(y=mean.size),col="red")+
  geom_line(aes(y=predict(lm(mean.size~mean.cap,data=flowclimate_data2))),col="red",linetype=2)+
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=predict(lm(opti.size~mean.cap,data=flowclimate_data2))),linetype=2)+
  theme_bw()+
  labs(title = "red : observed mean size, black : optimal size")+
  ylab("Size at flowering")+
  xlab("Mean number of capitulas produced")

```

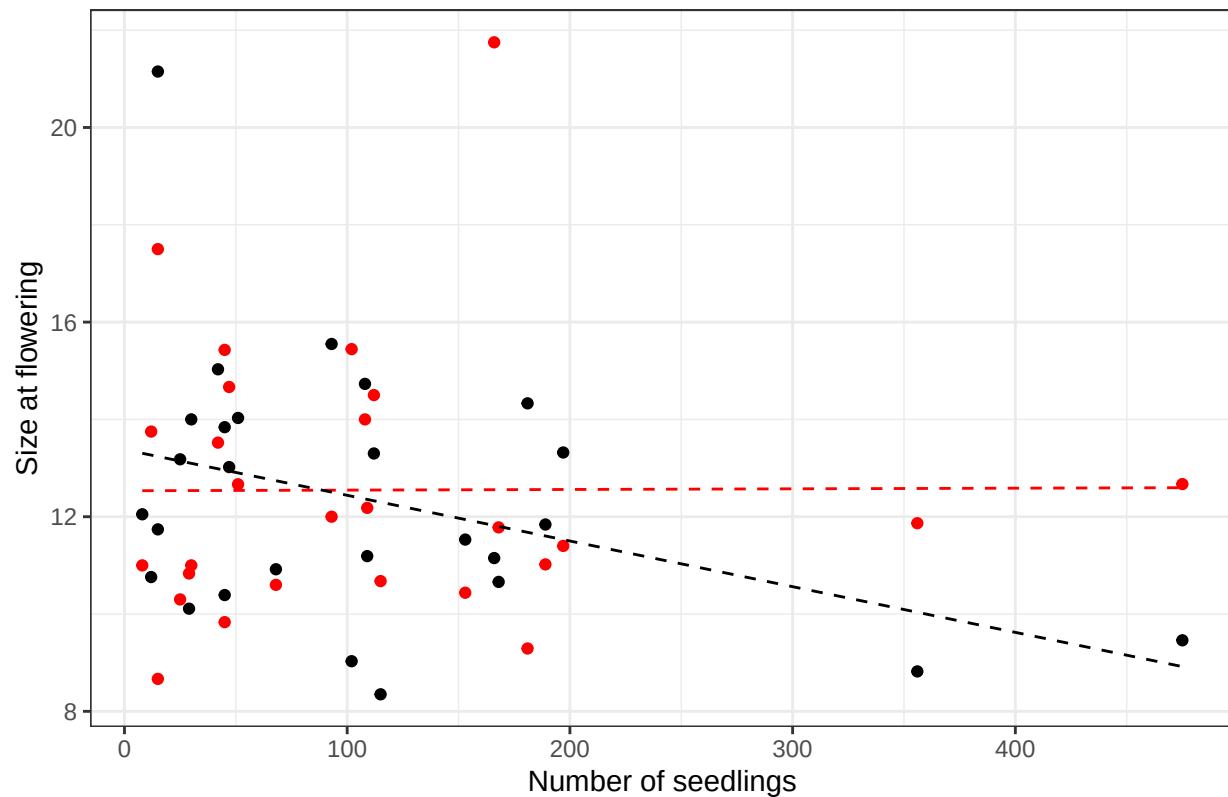


```

flowclimate_data2 %>%
  mutate(Seedlings = c(Seedlings[-1],NA)) %>%
  ggplot(aes(x=Seedlings))+
  geom_point(aes(y=mean.size),col="red")+
  geom_line(aes(y=c(predict(lm(mean.size~Seedlings,data=flowclimate_data2%>%
    mutate(Seedlings = c(Seedlings[-1],NA))))),NA)),col="red",linetype=2)+
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=c(predict(lm(opti.size~Seedlings,data=flowclimate_data2%>%
    mutate(Seedlings = c(Seedlings[-1],NA))))),NA)),linetype=2)+
  theme_bw()+
  labs(title = "red : observed mean size, black : optimal size")+
  ylab("Size at flowering")+
  xlab("Number of seedlings")

```

red : observed mean size, black : optimal size



Life history traits compared with climatic data

```
load("ModelsAIC")
load("IPMKernalYear_AIC")

annees <- 1995:2022
populations <- c("Po", "Au", "Pe", "E1", "E2", "Cr")
n.size <- 50
minsize <- 0.5
maxsize <- 25
maxsize1 <- 15
h <- (maxsize - minsize) / n.size
h1 <- (maxsize1 - minsize) / n.size
b = minsize + c(0:n.size) * h
#sizes for an individual of age a
#midpoint
ymid.a = 0.5 * (b[1:n.size] + b[2:(n.size + 1)])
#intervals
yI.a <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.a[i,1] <- b[i]
  yI.a[i,2] <- b[i+1]
}
```

```

b1 = minsize + c(0:n.size) * h1
#sizes for an individual of age 1
#midpoint
ymid.1 = 0.5 * (b1[1:n.size] + b1[2:(n.size + 1)])
#intervals
yI.1 <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.1[i,1] <- b1[i]
  yI.1[i,2] <- b1[i+1]
}

```

Survival probability

```

#Survival Probability for age 1
sx01 <- function(x,a,Pop,year) {
  fake_data <- expand.grid(
    year = year,
    Pop = Pop,
    Size0Mars = unique(x),
    Age = a)
  predi <- predict(Survglm11, fake_data, allow.new.levels = T)
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean)
  return(meanpredi$V1)
}

#Survival Probability for age > 1
sx02 <- function(x,a,Pop,year) {
  fake_data <- expand.grid(
    year = year,
    Pop = Pop,
    Size0Mars = unique(x),
    Age = a)
  predi <- predict(Survglm12, fake_data, allow.new.levels = T)
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean)
  return(meanpredi$V1)
}

```

```

mean.surv <- NULL
yr <- 1
for(yr in 1:length(annees)) {
  A <- eigen.analysis(IPMKernalYear_AIC[[yr]])
  N <- matrix(ncol = 8, nrow = 50)
  for (i in 1:8) {
    N[, i] <- A$stable.stage[(50 * (i - 1) + 1):(50 * i)]
  }
  W <- function(age) {
    return(N[(50 * (age - 1) + 1):(50 * age)])
  }
  mean.surv[yr] <- sum(sx01(ymid.1,1,Pop=populations,year=annees[yr])*W(1)/sum(W(1)))*sum(W(1))

  for (a in 2:8) {

```

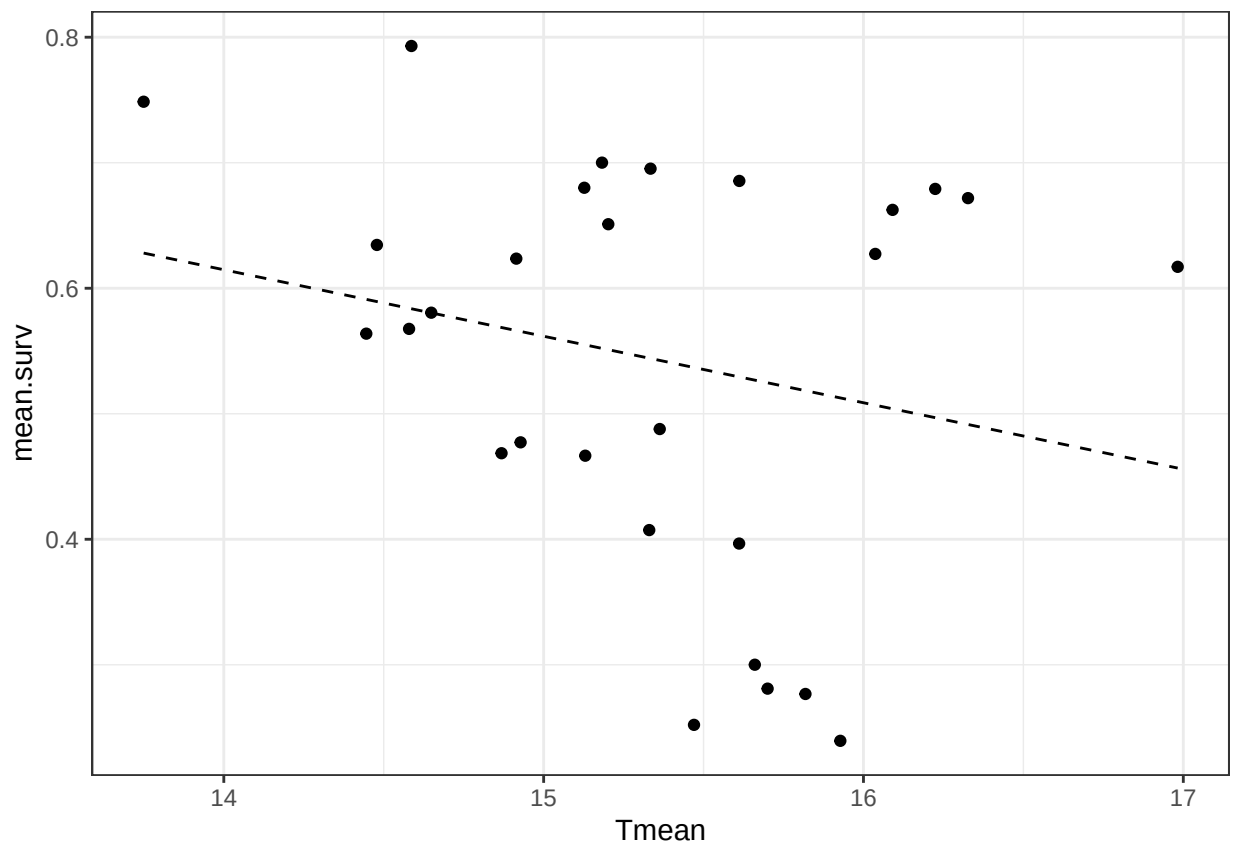
```

    mean.surv[yr] <- mean.surv[yr] + sum(sx02(ymid.a,a,Pop=populations,year=annees[yr])*W(a)/sum(W(a)))
  }
}

survpredi_data <- tibble(year=annees,
  mean.surv=mean.surv,
  Tmean=climate_data_mean$Tmean[-1])

survpredi_data %>%
  ggplot(aes(x = Tmean, y = mean.surv))+
  geom_point()+
  geom_line(aes(y=predict(lm(mean.surv~Tmean,data=survpredi_data))),linetype=2)+
  theme_bw()

```



The mean survival probability is negatively correlated with the temperature. A plant has less chance to survive if it's warmer, ok.

Flowering probability

```

flr0 <- function(x, a, Beta=0,Pop,year) {
  fake_data <- expand.grid(
    year = year,
    Pop = Pop,

```



```

Size0Mars = unique(x),
Age = a)
predi <- plogis(predict(Flowglm1, newdata = fake_data, allow.new.levels = T, type = "link") + Beta)
meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean)
return(meanpredi$V1)
}

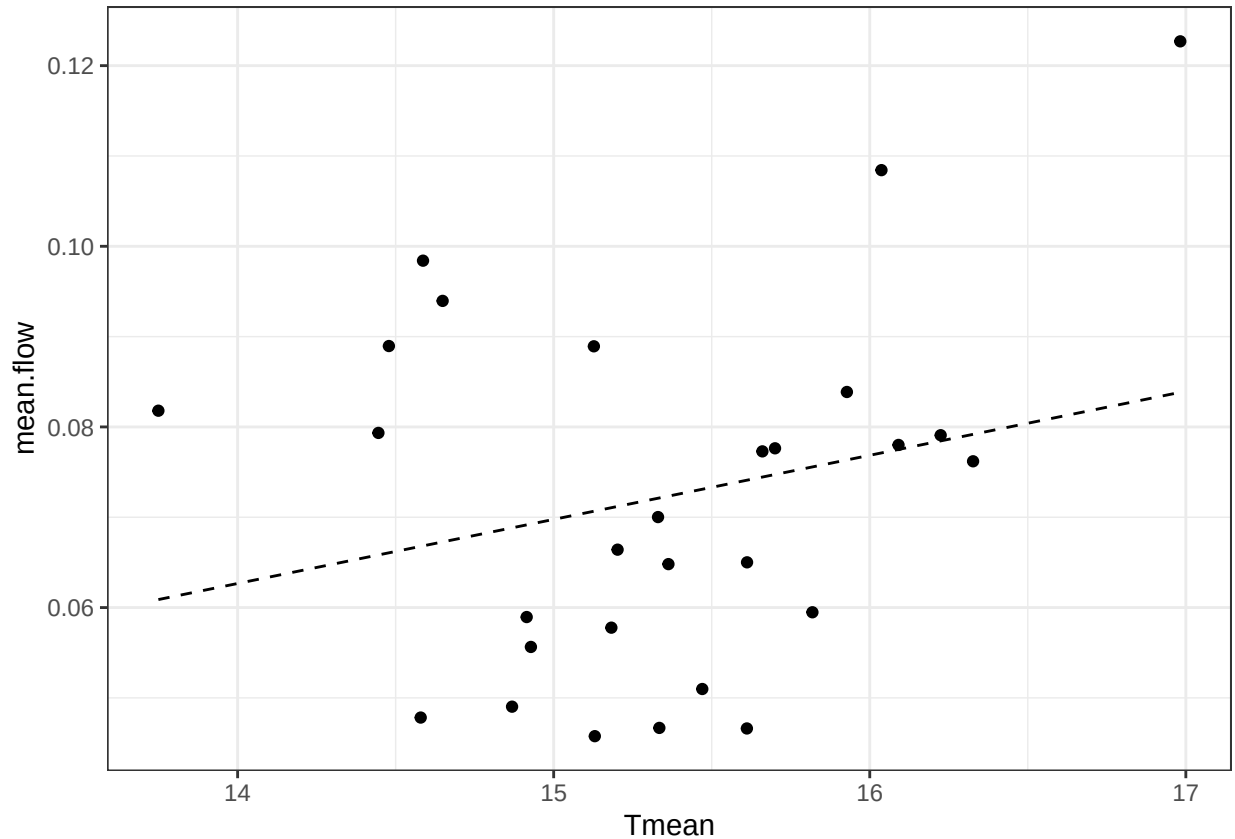
mean.flow <- NULL
yr <- 1
for(yr in 1:length(annees)) {
  A <- eigen.analysis(IPMKernalYear_AIC[[yr]])
  N <- matrix(ncol = 8, nrow = 50)
  for (i in 1:8) {
    N[, i] <- A$stable.stage[(50 * (i - 1) + 1):(50 * i)]
  }
  W <- function(age) {
    return(N[(50 * (age - 1) + 1):(50 * age)])
  }
  mean.flow[yr] <- sum(flR0(ymid.1,1,Pop=populations,year=annees[yr])*W(1)/sum(W(1)))*sum(W(1))

  for (a in 2:8) {
    mean.flow[yr] <- mean.flow[yr] + sum(flR0(ymid.a,a,Pop=populations,year=annees[yr])*W(a)/sum(W(a)))
  }
}

flowpredi_data <- tibble(year=annees,
                        mean.flow=mean.flow,
                        Tmean=climate_data_mean$Tmean[-1])

flowpredi_data %>%
  ggplot(aes(x = Tmean, y = mean.flow))+
  geom_point()+
  geom_line(aes(y=predict(lm(mean.flow~Tmean,data=flowpredi_data))),linetype=2)+
  theme_bw()

```



The mean flowering probability is positively correlated with temperature. If the year is warmer, a plant will flower now instead of wait the next year and risk to die during summer.

Fecundity

```
numbofcap <- function(x, a, year) {
  fake_data <- expand.grid(
    year = year,
    Size0Mars = unique(x),
    Age = a)
  predi <- exp(predict(Cptlg1m1, fake_data)) # prédiction du nombre de capitules produits pour iun indi
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean) #transforme le vecteur de prédiction e
  nbcap <- ifelse(meanpredi$V1>=0,meanpredi$V1,0)
  return(matrix(nbcap, length(x), length(x), byrow = T))
}
```

```
mean.capi <- NULL
yr <- 1
for(yr in 1:length(annees)) {
  A <- eigen.analysis(IPMKernalYear_AIC[[yr]])
  N <- matrix(ncol = 8, nrow = 50)
  for (i in 1:8) {
    N[, i] <- A$stable.stage[(50 * (i - 1) + 1):(50 * i)]
  }
}
```

```

W <- function(age) {
  return(N[(50 * (age - 1) + 1):(50 * age)])
}
mean.capi[yr] <- sum(flr0(ymid.1,1,Pop=populations,year=annees[yr])*numbofcap(ymid.1,1,year=annees[yr]))

for (a in 2:8) {
  mean.capi[yr] <- mean.capi[yr] + sum(flr0(ymid.a,a,Pop=populations,year=annees[yr])*numbofcap(ymid.a,a,year=annees[yr]))
}

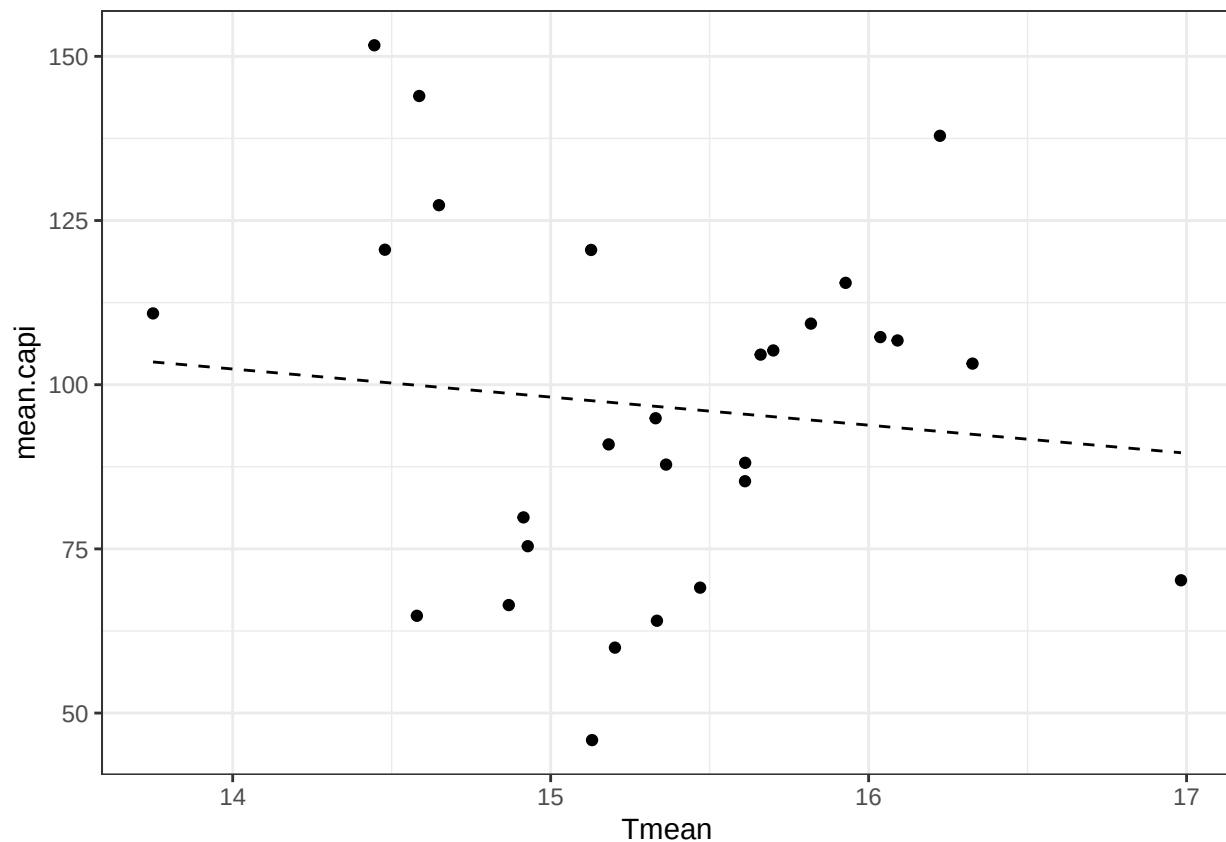
capipredi_data <- tibble(year=annees,
  mean.capi=mean.capi,
  Tmean=climate_data_mean$Tmean[-1])

```

```

capipredi_data %>%
  ggplot(aes(x = Tmean, y = mean.capi))+
  geom_point()+
  geom_line(aes(y=predict(lm(mean.capi~Tmean,data=capipredi_data))),linetype=2)+
  theme_bw()

```

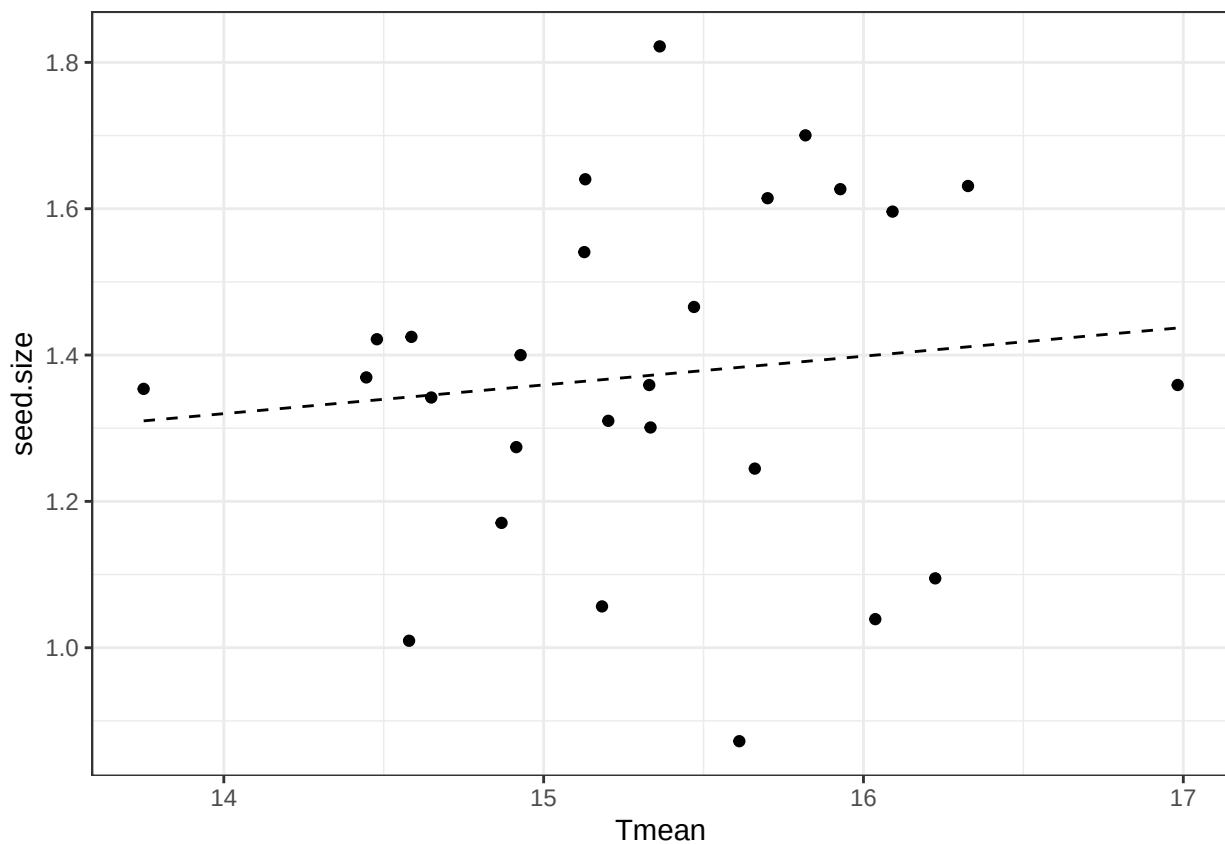


Seedling size

Effet de la température à t sur la taille des seedlings à t+1

```
seed.size <- NULL
for(yr in 1:length(annees)){
  seed.size[yr] <- mean(predict(Pltglm1, newdata = expand.grid(year=annees[yr],Pop=populations))[,1])
}
seedsizepredi_data <- tibble(year=annees,
                             Tmean=c(NA,climate_data_mean$Tmean[-c(1,28)]),
                             seed.size=seed.size)
```

```
seedsizepredi_data %>%
  filter(year!=1995) %>%
  ggplot(aes(x = Tmean, y = seed.size))+
  geom_point()+
  geom_line(aes(y=predict(lm(seed.size~Tmean,data=seedsizepredi_data))),linetype=2)+
  theme_bw()
```



Growth

```
Gyx3 <- function(ysup,yinf,x,a,Pop,year){
  sizepow <- 0.4343354
  yinfpow <- yinf**sizepow
  ysuppow <- ysup**sizepow
  yinfpow[1] <- -Inf
  ysuppow[length(ysuppow)] <- Inf

  fake_data <- expand.grid(
    year = year,
```

```

Pop = Pop,
Size0Mars = unique(x),
Age = a)
predi <- predict(Growthglm2, fake_data, allow.new.levels = T) #Prédit les taille~pow à t+1 possibles
meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean) #Calcule la moyenne des taille~t+1

resvar <- predict(Growthglm2$resid_fit, newdata = fake_data, allow.new.levels = T)
meanresvar <- aggregate(resvar[,1], list(fake_data$Size0Mars), mean)$x
cdf.sup <- mapply(pnorm, mean = meanpredi$V1, sd = sqrt(meanresvar), MoreArgs = list(q=ysuppow))
cdf.inf <- mapply(pnorm, mean = meanpredi$V1, sd = sqrt(meanresvar), MoreArgs = list(q=yinfpow))
return(cdf.sup - cdf.inf)
}

```

```

mean.size <- NULL
mean.size1 <- array(0,dim=length(annees))

for(yr in 1:length(annees)) {
  A <- eigen.analysis(IPMKernalYear_AIC[[yr]])
  N <- matrix(ncol = 8, nrow = 50)
  for (i in 1:8) {
    N[, i] <- A$stable.stage[(50 * (i - 1) + 1):(50 * i)]
  }
  W <- function(age) {
    return(N[(50 * (age - 1) + 1):(50 * age)])
  }

  G <- Gyx3(ysup=yI.a[,2], yinf=yI.a[,1], x=ymid.1, a=1, Pop=populations, year=annees[yr])
  for(i in 1:50){
    mean.size1[yr] <- mean.size1[yr] + ymid.a[i]*sum(G[i,]*W(1))
  }
  mean.size[yr] <- sum(ymid.1*W(1)/sum(W(1))*sum(W(1))

  for (a in 2:8) {
    mean.size[yr] <- mean.size[yr] + sum(ymid.a*W(a)/sum(W(a))) * sum(W(a))
    G <- Gyx3(ysup=yI.a[,2], yinf=yI.a[,1], x=ymid.a, a=a, Pop=populations, year=annees[yr])
    for(i in 1:50){
      mean.size1[yr] <- mean.size1[yr] + ymid.a[i]*sum(G[i,]*W(a))
    }
  }
}

growthpredi_data <- tibble(year=annees,
                           mean.size=mean.size,
                           mean.size1=mean.size1,
                           Tmean=climate_data_mean$Tmean[-1])

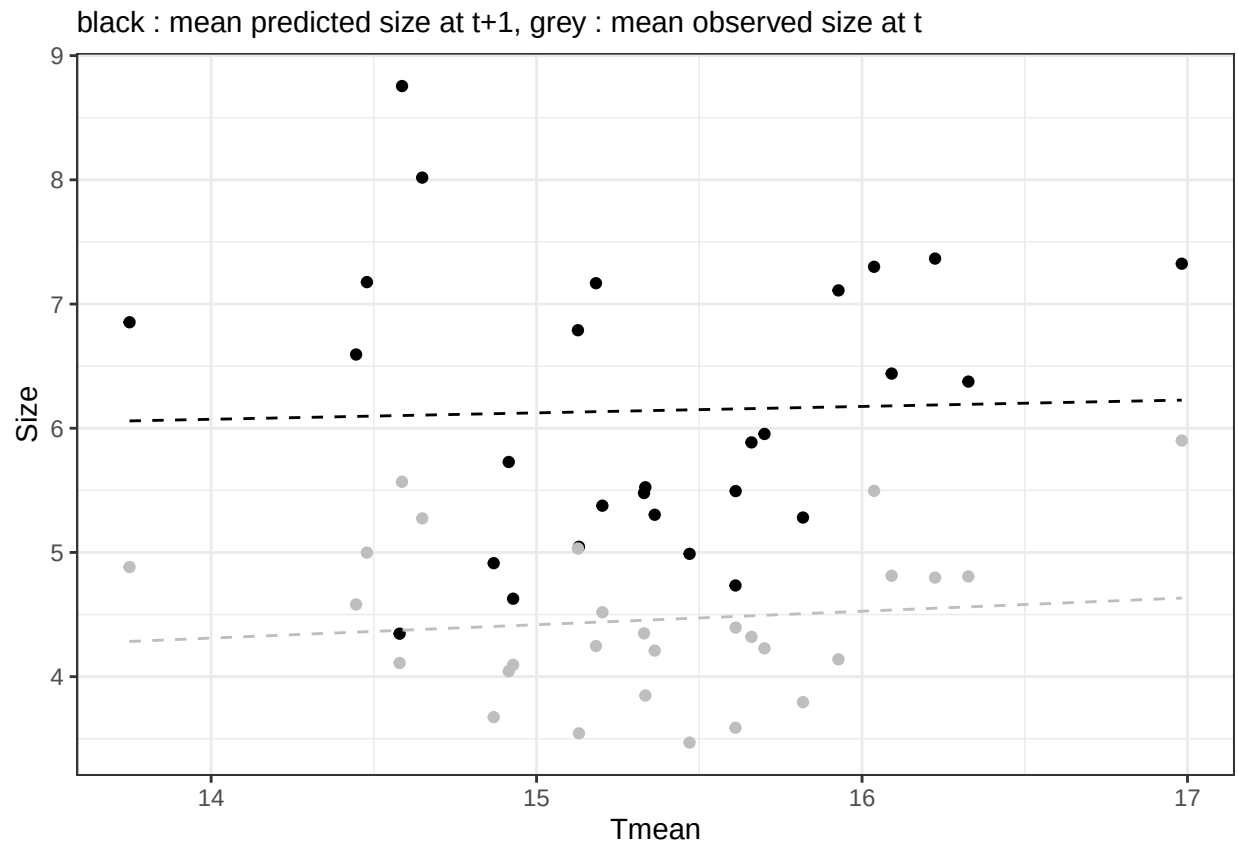
```

```

growthpredi_data %>%
  ggplot(aes(x = Tmean))+
  geom_point(aes(y = mean.size1))+
  geom_line(aes(y=predict(lm(mean.size1~Tmean,data=growthpredi_data))),linetype=2)+
  geom_point(aes(y = mean.size),col="green")+
  geom_line(aes(y=predict(lm(mean.size~Tmean,data=growthpredi_data))),linetype=2,col="green")+
  theme_bw()+

```

```
labs(subtitle = "black : mean predicted size at t+1, grey : mean observed size at t")+
ylab("Size")
```



Number of seedlings

```
flowclimate_data2 %>%
  mutate(Seedlings = c(NA,Seedlings[-1])) %>%
  ggplot(aes(x=Tmean,y=Seedlings))+
  geom_point()+
  geom_line(aes(y=predict(lm(Seedlings~Tmean,data=flowclimate_data2))),linetype=2)+
  theme_bw()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

